

Exchange Rates for Finite Map Signatures

A Simplified Thermodynamic Complexity

Artem Vorozhtsov

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Abstract

We associate to a surjective map between finite sets the multiset of its fiber cardinalities. Cartesian products turn these signatures into a simple asymptotic resource theory. Its exchange rate $C(g \mid f)$ is the largest asymptotic number of copies of f implementable per copy of g . Power sums compute this operationally defined limit, while Gibbs energy–entropy curves provide a geometric picture. The comparison obtained by putting the two directed rates side by side is not transitive: the three signatures $\{5, 3\}$, $\{3, 1, 1\}$, and $\{6, 1\}$ form a strict cycle. This is a concrete monoidal resource theory of finite deterministic computational tasks: its one-shot convertibility resembles strong Weihrauch reducibility, while its asymptotic exchange rate is a finite-map analogue of tensor relative exponents and Strassen’s asymptotic spectrum.

1 Signatures and implementation

Let $f : X \rightarrow Y$ be onto and enumerate its output as $Y = \{y_1, \dots, y_r\}$. The *fiber* over y_i is the inverse image

$$f^{-1}(y_i) = \{x \in X : f(x) = y_i\}, \quad n_i = |f^{-1}(y_i)|.$$

Thus n_i is the number of inputs mapped by f to the output y_i . After ordering these fiber sizes by decreasing value, the *signature* of f is

$$\sigma(f) = (n_1, \dots, n_r), \quad n_1 \geq \dots \geq n_r \geq 1, \quad r = |Y|.$$

Only the signature will matter below, so maps and their signatures are denoted by the same letters.

Definition 1 (Implemented by). Let $a = (a_1, \dots, a_r)$ and $b = (b_1, \dots, b_s)$ be decreasing signatures. We say that a is *implemented by* b , and write $a \preceq_{\text{impl}} b$, when $r \leq s$ and

$$a_i \leq b_i \quad (1 \leq i \leq r).$$

For maps, this means that there is an injective input relabeling h_{in} and a bijection from the used output fibers to the output of f such that

$$f = h_{\text{out}} \circ g \circ h_{\text{in}}.$$

More precisely, put

$$X'_g = h_{\text{in}}(X_f), \quad Y'_g = g(X'_g).$$

The two inclusions on the right identify the parts of X_g and Y_g actually used by the implementation, and the following diagram commutes:

$$\begin{array}{ccccc}
X_f & \xrightarrow[\cong]{h_{\text{in}}} & X'_g & \hookrightarrow & X_g \\
f \downarrow & & \downarrow g|_{X'_g} & & \downarrow g \\
Y_f & \xleftarrow[\cong]{h_{\text{out}}} & Y'_g & \hookrightarrow & Y_g.
\end{array}$$

Unused fibers of g may be discarded. The componentwise condition above is exactly the condition that every fiber of f can be injected into its assigned fiber of g .

The Cartesian product $a \otimes b$ has all products $a_i b_j$ as its fiber sizes, with multiplicity. We write $a^{\otimes k}$ for the k -fold product.

$$a \otimes b = \text{sort}\{a_i b_j : 1 \leq i \leq r, 1 \leq j \leq s\}.$$

There is also a natural sum. The disjoint union of maps

$$f \sqcup g : X_f \sqcup X_g \longrightarrow Y_f \sqcup Y_g$$

has as its signature the sorted concatenation of the two signatures; we write this as

$$a + b = \text{sort}(a_1, \dots, a_r, b_1, \dots, b_s).$$

Both operations respect implementation, and Cartesian product distributes over this sum:

$$a \preceq_{\text{impl}} b, c \preceq_{\text{impl}} d \implies a + c \preceq_{\text{impl}} b + d, \quad a \otimes c \preceq_{\text{impl}} b \otimes d, \quad a \otimes (b + c) = (a \otimes b) + (a \otimes c).$$

Thus signatures form the nonzero part of an ordered commutative semiring (adjoining the empty signature gives the additive zero), with multiplicative unit $\{1\}$. The sum represents a disjoint, classically labelled alternative, whereas $\otimes$ represents independent parallel composition. Only $\otimes$ is needed in the exchange-rate definition below.

Definition 2 (Exchange rate). For signatures f and g , set

$$k_{\max}^{g \leftarrow f}(n) = \max \left\{ k \geq 0 : f^{\otimes k} \preceq_{\text{impl}} g^{\otimes n} \right\}.$$

The exchange rate, with the implementer written first, is

$$C(g \mid f) = \lim_{n \rightarrow \infty} \frac{k_{\max}^{g \leftarrow f}(n)}{n}.$$

The sequence $k_{\max}$ is superadditive: implementations for n and m copies can be tensored. Fekete's lemma therefore gives the limit (possibly $+\infty$) and identifies it with the corresponding supremum. Thus the limit is the definition; the infimum below is a computational characterization.

2 Related frameworks

The present construction is a concrete semiring resource theory: signatures are resources, implementation is free conversion, disjoint union supplies addition, and Cartesian product combines independent resources. General resource theories formulate convertibility from freely available

processes and explicitly ask for rates between many copies, while ordered commutative monoids and semirings provide an algebraic treatment of conversion, combination, and asymptotic rates (see, e.g., [1, 2]).

At the one-shot level, the factorization

$$f = h_{\text{out}} \circ g \circ h_{\text{in}}$$

resembles strong Weihrauch reducibility of computational problems, in which a single use of g is surrounded by computable preprocessing and postprocessing (see [3]). Weihrauch theory also has a parallel product. The important distinction is that our finite processors are injective or bijective on the used sets; arbitrary computable processors would forget the fiber capacities that this paper retains.

An even closer asymptotic precedent comes from algebraic complexity. Tensors encode bilinear computational tasks, restriction converts one tensor to another by linear maps, and tensor product combines independent copies. Strassen's asymptotic spectrum characterizes large tensor powers by multiplicative monotones (see [4]). The relative exponent

$$\omega(t, s) = \lim_{n \rightarrow \infty} \frac{1}{n} \min\{m : t^{\otimes m} \text{ restricts to } s^{\otimes n}\}$$

is the resource cost per target copy and has been used to quantify irreversibility in matrix-multiplication methods (see [5]). With the direction reversed, it is the reciprocal convention to $C(t \mid s)$. In our setting the multiplicative quantities $Z_a(\beta)$, or their additive logarithms, play the role of spectral monotones. The resulting infimum over power sums is also close to the exact asymptotic-majorization rate, expressed as a minimum of ratios of Rényi entropies (see [6]).

Finally, channel simulation surrounds repeated uses of a resource channel by an encoder and decoder in order to reproduce a target channel. For point-to-point discrete memoryless channels with common-randomness assistance, the first-order interconversion rate is the ratio of channel capacities (see [7]). Appendix A explains why its unrestricted finite-map variant has exactly the corresponding logarithmic output-cardinality form.

3 Power sums and Gibbs curves

For a signature $a = (a_i)$, define its power-sum partition function

$$Z_a(\beta) = \sum_i a_i^\beta, \quad \beta \geq 0,$$

where $\beta = 1/T$ is inverse physical temperature. Throughout this paper, T denotes temperature and never inverse temperature. The sum and product of signatures become the ordinary sum and product of partition functions:

$$Z_{a+b}(\beta) = Z_a(\beta) + Z_b(\beta), \quad Z_{a \otimes b}(\beta) = Z_a(\beta) Z_b(\beta).$$

The fibers of the k -fold Cartesian power $a^{\otimes k}$ are indexed by $(i_1, \dots, i_k)$ and have sizes $a_{i_1} \cdots a_{i_k}$. Consequently,

$$\begin{aligned} Z_{a^{\otimes k}}(\beta) &= \sum_{i_1, \dots, i_k} (a_{i_1} \cdots a_{i_k})^\beta \\ &= \left(\sum_i a_i^\beta \right)^k = Z_a(\beta)^k, \quad \log Z_{a^{\otimes k}}(\beta) = k \log Z_a(\beta). \end{aligned} \tag{1}$$

If $f^{\otimes k} \preceq_{\text{impl}} g^{\otimes n}$, applying the power sum at every $\beta \geq 0$ gives

$$Z_f(\beta)^k \leq Z_g(\beta)^n, \quad \frac{k}{n} \leq \frac{\log Z_g(\beta)}{\log Z_f(\beta)}.$$

Thus Cartesian products become ordinary scalar multiplication after taking $\log Z$. Passing to the asymptotic optimum gives the upper bound in the following formula.

The asymptotic comparison is governed by

$$C(g \mid f) = \inf_{\beta \geq 0} \frac{\log Z_g(\beta)}{\log Z_f(\beta)}, \quad (2)$$

together with the endpoint

$$\lim_{\beta \rightarrow \infty} \frac{\log Z_g(\beta)}{\log Z_f(\beta)} = \frac{\log \max_j g_j}{\log \max_i f_i}.$$

Necessity follows by applying every power sum to an implementation. Asymptotic sufficiency follows by sorting the product fibers and applying the usual exponential, or large-deviation, estimates to their ranks. In this way all componentwise constraints are represented by the one-parameter family $Z_a(\beta)$.

Put

$$\epsilon_i = -\log a_i, \quad p_i(T) = \frac{e^{-\epsilon_i/T}}{\sum_j e^{-\epsilon_j/T}} = \frac{a_i^{1/T}}{Z_a(1/T)}.$$

These are the Gibbs probabilities at physical temperature T . Their mean energy and entropy are

$$E_a(T) = \sum_i p_i(T) \epsilon_i, \quad H_a(T) = -\sum_i p_i(T) \log p_i(T).$$

Writing $\beta = 1/T$, they satisfy $\log Z_a(\beta) = H_a(T) - E_a(T)/T$. Moreover,

$$H_{a^{\otimes k}}(T) = kH_a(T), \quad E_{a^{\otimes k}}(T) = kE_a(T),$$

because the Gibbs distribution of the Cartesian power is the product distribution. Eliminating T gives

$$E_{a^{\otimes k}}(H) = kE_a(H/k),$$

so taking Cartesian powers is a homothety by k in the (E, H) -plane.

Geometrically, $\log Z_a(\beta) = H_a(T) - \beta E_a(T)$ is the offset of the supporting line with normal $(-\beta, 1)$ to the convex Gibbs region $\mathcal{R}_a$. Therefore formula (2) is equivalently

$$C(g \mid f) = \sup\{c \geq 0 : c \mathcal{R}_f \subseteq \mathcal{R}_g\}.$$

In particular, $C(g \mid f)\Gamma_f$ fits in the convex Gibbs region bounded by Γ_g . Equivalently, the tight homothetic copy of g embracing f is $C(g \mid f)^{-1}\Gamma_g$.

4 The pair $\{2, 2\}$ and $\{3, 1\}$

Let $g = \{2, 2\}$ and $f = \{3, 1\}$. Every fiber of $g^{\otimes n}$ has size 2^n . The largest fiber of $f^{\otimes k}$ has size 3^k , and the remaining rank constraint is weaker at the optimum. Hence

$$C(\{2, 2\} \mid \{3, 1\}) = \frac{\log 2}{\log 3} = 0.6309297536 \dots$$

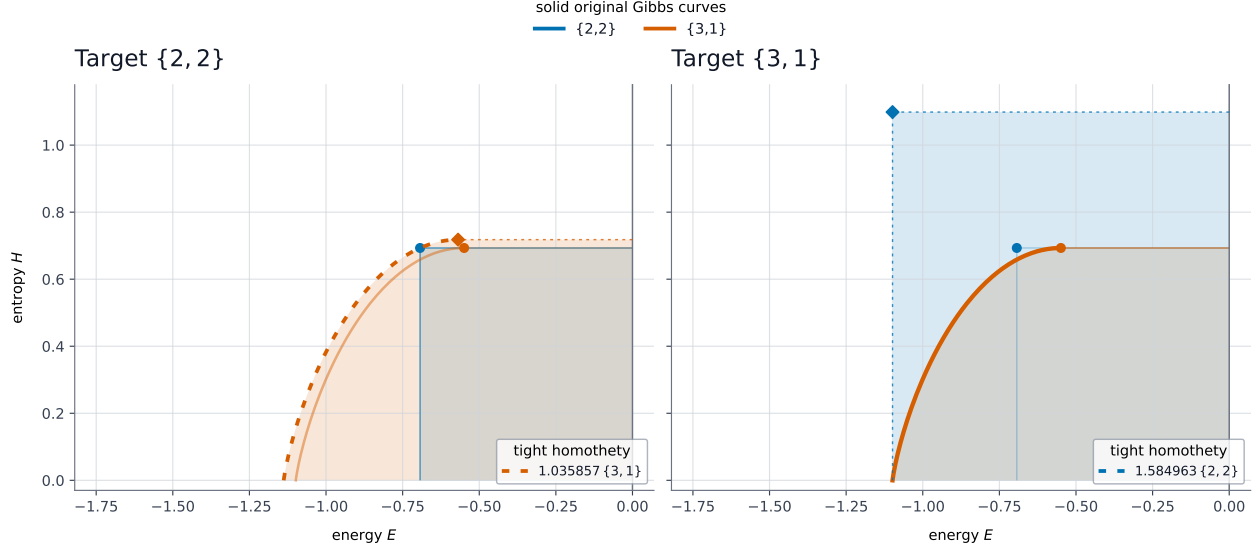


Figure 1: The two directed homotheties for $\{2, 2\}$ and $\{3, 1\}$. Thin horizontal lines close the $T = \infty$ endpoints. Vertical lines close only $T = 0$ endpoints with $H(0) > 0$. Both horizontal and vertical closures are dotted for homothetically scaled boundaries; translucent fills extend to $E = 0$. The factors are $1/C(\{3, 1\} \mid \{2, 2\}) = 1.03585676\dots$ on the left and $1/C(\{2, 2\} \mid \{3, 1\}) = \log 3 / \log 2$ on the right.

The reverse direction is limited at a finite temperature:

$$\begin{aligned}
 C(\{3, 1\} \mid \{2, 2\}) &= \min_{\beta \geq 0} \frac{\log(3^\beta + 1)}{(\beta + 1) \log 2} \\
 &= 0.9653844417\dots, \quad \beta_* = 0.4036799030\dots, \quad T_* = \beta_*^{-1} = 2.47721027\dots
 \end{aligned}$$

The large asymmetry is visible in Figure 1. Solid lines are the original Gibbs boundaries. In each panel the other signature is enlarged by the reciprocal directed rate; only this resulting tight boundary is dotted. The thin horizontal line joins the $T = \infty$ endpoint to $E = 0$. When the $T = 0$ endpoint has residual entropy $H(0) > 0$, a thin vertical line joins that endpoint to $H = 0$; no vertical line is needed when $H(0) = 0$. Endpoint closures of homothetically scaled boundaries are thin and dotted, matching the scaled curve; endpoint closures of original boundaries are thin and solid. The 85%-transparent fills extend horizontally from each boundary to $E = 0$, displaying the closed convex Gibbs regions. Since $\{2, 2\}$ has two equal energy levels, its canonical Gibbs boundary itself collapses to the single point $(E, H) = (-\log 2, \log 2)$.

5 A two-sided finite-temperature tangency

Finite-temperature contact can determine both directed rates. Consider

$$a = \{6, 5, 2, 1\}, \quad b = \{6, 4, 3, 2\}.$$

Because a and b have the same length and the same largest fiber, both endpoint comparisons are equal to one:

$$\frac{\log Z_a(0)}{\log Z_b(0)} = 1, \quad \lim_{\beta \rightarrow \infty} \frac{\log Z_a(\beta)}{\log Z_b(\beta)} = \frac{\log 6}{\log 6} = 1.$$

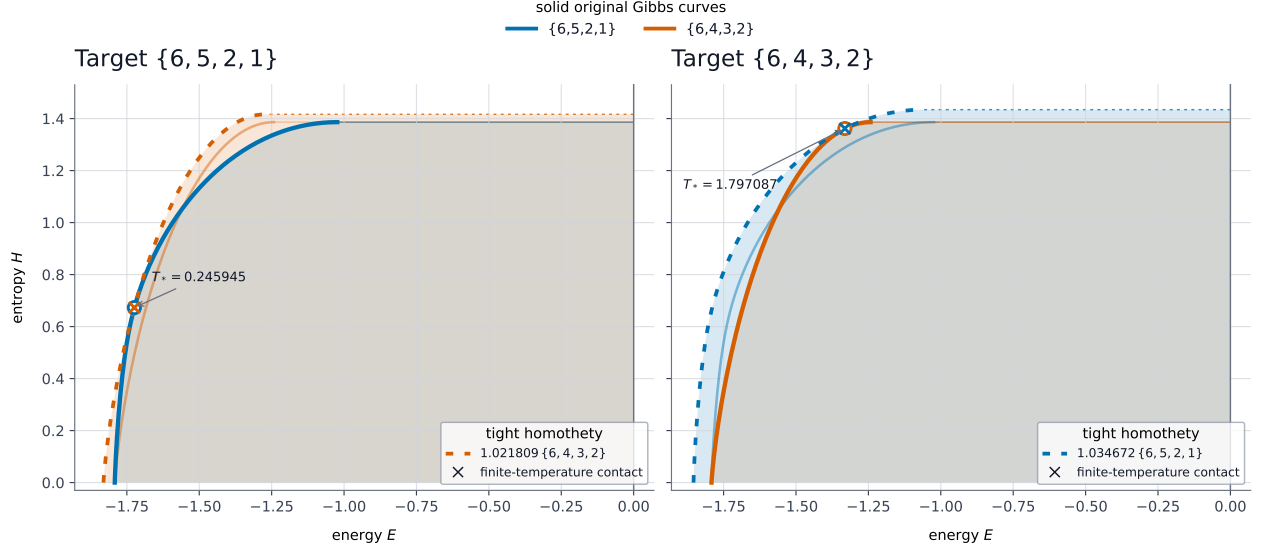


Figure 2: Two-sided finite-temperature tangency. The left panel scales b by $1/C(b | a)$ to embrace a ; the right panel scales a by $1/C(a | b)$ to embrace b . Crosses mark the interior contact points. Thin dotted horizontal lines close the scaled regions at their $T = \infty$ endpoints; translucent fills extend to $E = 0$.

Nevertheless, the two directed minima are strictly smaller than one and occur at finite positive temperatures:

Direction	C	T_*	$\beta_* = 1/T_*$
$C(a b)$	0.9664896992	1.7970865612	0.5564562229
$C(b a)$	0.9786567233	0.2459449336	4.0659508025

At an interior minimizer of $R(\beta) = \log Z_g(\beta) / \log Z_f(\beta)$, the equation $R'(\beta_*) = 0$, together with $\log Z = H - \beta E$, gives

$$E_g(T_*) = C(g | f)E_f(T_*), \quad H_g(T_*) = C(g | f)H_f(T_*).$$

Thus the scaled Gibbs boundary and the target boundary meet at the same point. Moreover

$$\frac{dH}{dE} = \frac{1}{T_*} = \beta_*,$$

which is invariant under homothety, so their tangent lines also agree. Both contacts in Figure 2 are therefore genuine finite-temperature tangencies rather than endpoint contacts.

6 A cyclic strict comparison

Definition 3. For signatures a, b , define

$$a \prec b \iff C(a | b) < C(b | a).$$

This is a natural pairwise comparison: b is declared more complex when it converts into a more efficiently than a converts into b . Nevertheless, it is not a strict linear order.

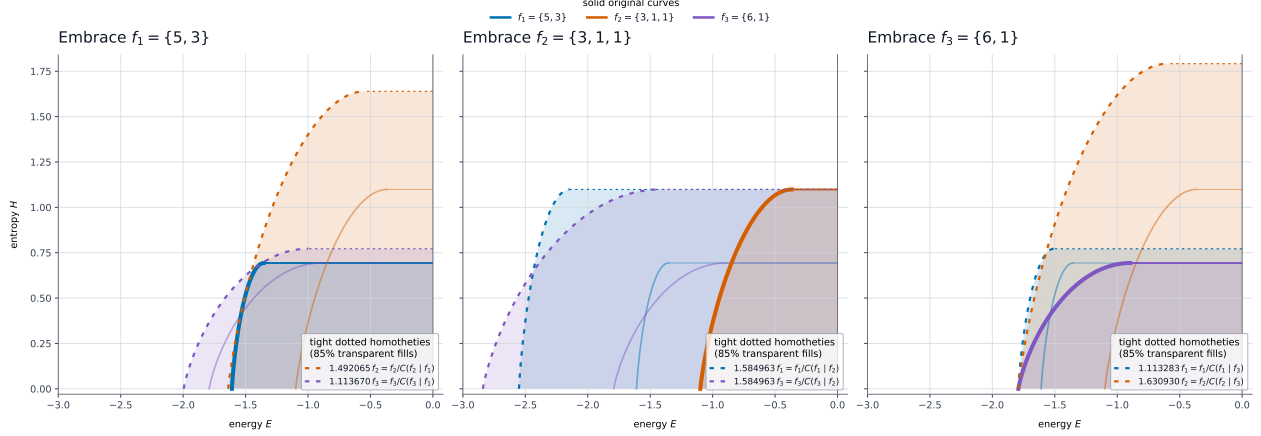


Figure 3: The cyclic example. Solid and dotted boundaries are original and scaled, respectively. Horizontal $T = \infty$ closures and 85%-transparent fills extending to $E = 0$ display target containment.

Set

$$f_1 = \{5, 3\}, \quad f_2 = \{3, 1, 1\}, \quad f_3 = \{6, 1\}.$$

Equation (2) gives the comparisons in Table 1.

Table 1: Directed rates producing a strict cycle.

Pair	Forward rate	Reverse rate	Conclusion
f_1, f_2	$C(f_1 f_2) = 0.630929754$	$C(f_2 f_1) = 0.670211928$	$f_1 \prec f_2$
f_2, f_3	$C(f_2 f_3) = 0.613147193$	$C(f_3 f_2) = 0.630929754$	$f_2 \prec f_3$
f_3, f_1	$C(f_3 f_1) = 0.897931851$	$C(f_1 f_3) = 0.898244402$	$f_3 \prec f_1$

Thus

$$f_1 \prec f_2, \quad f_2 \prec f_3, \quad f_3 \prec f_1.$$

A strict linear order must be transitive, so this three-cycle proves that $\prec$ is not a linear order. Figure 3 shows all three pairwise comparisons geometrically. In the panel targeting f_i , the two dotted curves are $C(f_j | f_i)^{-1} \Gamma_{f_j}$; each is the tight homothetic boundary embracing Γ_{f_i} .

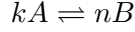
7 A thermodynamic analogy and its limits

There is an appealing physical reading of the construction. Regard a signature $A = (a_i)$ as one idealized particle whose internal energy levels are $\epsilon_i = -\log a_i$. The Cartesian power $A^{\otimes k}$ is then a system of k independent particles of type A . Likewise, a second signature B represents a second particle type. At an interior contact of their homothetically scaled Gibbs boundaries,

$$\frac{dH_A}{dE} = \frac{dH_B}{dE} = \frac{1}{T},$$

so the active comparison has a common temperature. This makes the finite-temperature tangencies above analogous to reversible contact between two gases of particles.

The analogy must not be read as a law of spontaneous reaction. The relation $\preccurlyeq_{\text{impl}}$ is a combinatorial possibility, not a microscopic dynamics, and C is a conversion capacity rather than a chemical affinity. For a putative conversion



put $r = k/n$. The definitions give the two distinct conditions

$$B^{\otimes n} \longrightarrow A^{\otimes k} \iff r \leq C(B \mid A) \quad \text{in the asymptotic closure,} \quad (3)$$

$$A^{\otimes k} \longrightarrow B^{\otimes n} \iff r \geq \frac{1}{C(A \mid B)} \quad \text{in the asymptotic closure.} \quad (4)$$

Strict inequalities guarantee the corresponding direction for sufficiently large block sizes; an exact boundary ratio need not be attained at finite size. Thus a single inequality involving $C(A \mid B)$ does not choose the direction. In fact,

$$C(A \mid B)C(B \mid A) \leq 1,$$

so the two thresholds form a conversion window

$$C(B \mid A) \leq r \leq \frac{1}{C(A \mid B)}.$$

Away from equality there can be ratios in this window for which neither direction realizes that exact stoichiometry. The width of the window is an irreversibility loss, analogous to a bid–ask spread. When the endpoints coincide, the comparison is asymptotically reversible. An interior tangency identifies the Gibbs temperature at which the corresponding supporting constraint is tight.

For comparison, genuine chemical equilibrium at fixed temperature and pressure would require zero reaction affinity,

$$\mathcal{A} = k\mu_A - n\mu_B = 0,$$

in addition to equal temperature. Here μ_A, μ_B are chemical potentials. This condition is not part of the definition of implementation. Consequently, the rate inequalities should be called *allowed conversion directions*, not spontaneous reaction directions.

The strict three-cycle suggests a sharper, explicitly nonequilibrium analogy. For a strict comparison arrow $a \xrightarrow{s} b$, define the dimensionless *complexity affinity*

$$\alpha(a \rightarrow b) = \log \frac{C(a \mid b)}{C(b \mid a)} > 0.$$

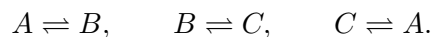
For the cycle in Section 6, whose arrow orientation is

$$f_1 \xrightarrow{s} f_3 \xrightarrow{s} f_2 \xrightarrow{s} f_1,$$

the total complexity-cycle affinity is

$$\begin{aligned} \mathcal{A}_{\text{cyc}} &= \alpha(f_1 \rightarrow f_3) + \alpha(f_3 \rightarrow f_2) + \alpha(f_2 \rightarrow f_1) \\ &= \log \frac{0.898244402}{0.897931851} + \log \frac{0.630929754}{0.613147193} + \log \frac{0.670211928}{0.630929754} \\ &= 0.089337, \quad e^{\mathcal{A}_{\text{cyc}}} = 1.09345. \end{aligned}$$

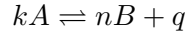
If α is formally read as a logarithmic forward–reverse reaction bias, this resembles a preferred cycle in a three-component reaction network,



Such a cycle cannot describe spontaneous circulation in a closed equilibrium gas. Reaction affinity is the negative derivative of Gibbs energy with respect to reaction extent (see [8]); because Gibbs energy is a state function, the total affinity around a closed equilibrium cycle vanishes. Equivalently, detailed balance imposes the Wegscheider cycle condition, so all three directions cannot simultaneously have strictly positive equilibrium affinity. A nonzero cycle affinity instead belongs naturally to an open system driven by chemostats, fuel, radiation, or other reservoirs. Open chemical reaction networks with emergent cycles and nonvanishing affinities are developed by Polettini and Esposito [9] and Rao and Esposito [10]; Schmiedl and Seifert [11] give a stochastic-thermodynamic three-species cyclic example.

This identification remains structural rather than physical. The quantities $C(a | b)$ are conversion capacities, not kinetic rate constants; the three pairwise infima may occur at different temperatures; and the signature model supplies neither reaction kinetics nor conservation laws, stoichiometric closure, or external work accounting. Hence a strict comparison cycle is a *candidate nonequilibrium reaction cycle*: it gives an orientation and a formal cycle affinity, but by itself predicts neither a sustained current nor chemical oscillations.

A constant reaction-energy term can nevertheless be represented cleanly. For



shift every level of B by q/n :

$$\tilde{\epsilon}_{B,j} = \epsilon_{B,j} + \frac{q}{n}.$$

Then

$$\tilde{E}_B(H) = E_B(H) + \frac{q}{n}, \quad \tilde{Z}_B(T) = e^{-q/(nT)} Z_B(T),$$

while $H_B(T)$ and the tangent slope $dH_B/dE = 1/T$ are unchanged. Hence the n -particle energy region moves horizontally by q . This bookkeeping is valid when q is a fixed energy offset carried by an explicit work or storage degree of freedom. Heat passed to a bath also carries entropy and cannot, in general, be modeled by a horizontal shift alone.

Several physical effects are deliberately absent: interactions, quantum symmetrization, translational and volume degrees of freedom, mixing entropy between particle types, conservation laws, reaction kinetics, activation barriers, and coupling to a heat or particle reservoir. The logarithmic energies are dimensionless ($k_B = 1$), and relative energy offsets between particle types must be fixed because a conversion may change particle number. Subject to these simplifications, the construction suggests a broader question: use homothetic containment of Gibbs regions to compare the complexity of general Gibbs systems.

The supporting function

$$\Psi_A(T) = \log Z_A(T) = -\frac{F_A(T)}{T}$$

is the Massieu potential, or free entropy. Thus

$$C(A | B) = \inf_{T>0} \frac{\Psi_A(T)}{\Psi_B(T)}$$

is the largest factor c for which $\Psi_A(T) \geq c\Psi_B(T)$ simultaneously at every temperature. It may be read as an all-temperature, or worst-temperature, conversion rate; an interior minimizing T_* is the bottleneck temperature. The closest established framework is the resource theory without a fixed background temperature developed by Sparaciari, Oppenheim, and Fritz (see [12]). There, asymptotically many noninteracting systems are classified by energy and entropy, and conversion

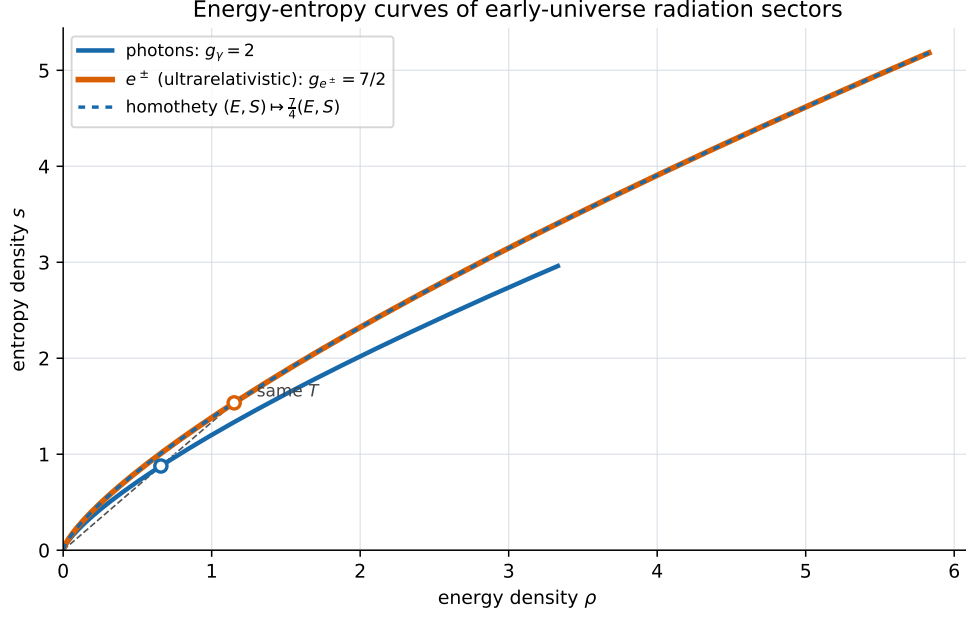


Figure 4: A real exact homothety in the ultrarelativistic limit. At the same temperature, the electron–positron sector has $7/4$ times the energy and entropy densities of the photon sector. The dotted homothetic photon curve coincides with the electron–positron curve.

rates are read from a two-dimensional energy–entropy diagram. Our construction specializes this idea to complete Gibbs boundaries and requires the comparison at all temperatures rather than relative to one prescribed heat bath.

A simple real system realizes the homothety exactly. For an ultrarelativistic equilibrium sector X with effective number g_X of internal degrees of freedom, in natural units and fixed volume,

$$E_X(T) = \frac{\pi^2}{30} g_X V T^4, \quad H_X(T) = \frac{2\pi^2}{45} g_X V T^3.$$

Consequently, two such sectors A, B obey

$$E_A(T) = \frac{g_A}{g_B} E_B(T), \quad H_A(T) = \frac{g_A}{g_B} H_B(T)$$

at every temperature: their entire energy–entropy boundaries are homothetic, with $C = g_A/g_B$. Photons have $g_\gamma = 2$, whereas an ultrarelativistic electron–positron sector has the fermionic effective value $g_{e^\pm} = (7/8) \times 4 = 7/2$. Hence its factor relative to photons is $7/4$, as shown in Figure 4. The early-universe reaction $e^- + e^+ \rightleftharpoons 2\gamma$ maintained thermal and chemical equilibrium in the relativistic regime; when the temperature fell below the electron mass, annihilation transferred entropy to the photon bath. Detailed calculations confirm that most annihilation occurred near equilibrium and that the process affects big-bang nucleosynthesis and the effective relativistic degrees of freedom (see [13]).

For arbitrary Gibbs systems this should initially be viewed as a proposed complexity comparison: unlike finite-map signatures, its operational interpretation and the treatment of infinite spectra require separate analysis.

8 Conclusion

Fiber signatures retain only a very small part of a finite map, yet their Cartesian products already produce a nontrivial asymptotic resource theory. The operational rate $C(g \mid f)$, the power-sum formula, and the energy–entropy homotheties are three views of the same comparison. The example above also warns against compressing the two directed rates into a single global ranking: pairwise exchange advantage can be cyclic. The thermodynamic analogy suggests using the same Gibbs-region comparison for broader systems while keeping conversion capacity distinct from physical reaction affinity.

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A Unrestricted input and output maps

Suppose that the maps h_{in} and h_{out} in the definition of implementation are allowed to be arbitrary. Then b can be implemented by a exactly when

$$|Y_b| \leq |Y_a|.$$

Indeed, arbitrary h_{in} may send every element of a fiber of b to one chosen input of a , so the fiber sizes impose no constraint. Conversely, a deterministic h_{out} cannot produce more distinct outputs than a has.

For Cartesian powers the condition is therefore

$$|Y_b|^k \leq |Y_a|^n, \quad k_{\max}(n) = \left\lfloor n \frac{\log |Y_a|}{\log |Y_b|} \right\rfloor.$$

Consequently, when $|Y_a|, |Y_b| > 1$,

$$C(a | b) = \frac{\log |Y_a|}{\log |Y_b|}.$$

This is the same first-order form as assisted point-to-point channel interconversion [7]: a surjective deterministic channel with output set Y has capacity $\log |Y|$, so the displayed rate is the ratio of the two channel capacities. The unrestricted variant therefore sees only ordinary communication capacity, whereas the injective and bijective processors in the main theory retain the individual fiber sizes.

Thus $C(a | b)C(b | a) = 1$, and the comparison from Definition 3 reduces to

$$a \prec b \iff |Y_a| < |Y_b|.$$

It is a total preorder on signatures, with equal-size output sets tied, and becomes a linear order after identifying signatures having the same $|Y|$. In partition-function language only the endpoint $Z_a(0) = |Y_a|$ remains; the fiber sizes and the rest of the Gibbs curve play no role. If $|Y_b| = 1$, then $C(a | b) = +\infty$; if $|Y_a| = 1 < |Y_b|$, then $C(a | b) = 0$.

B The first 69 non-special signatures

Singleton signatures $\{N\}$ and all-ones signatures $\{1, \dots, 1\}$ form exceptional infinite families and are omitted from this finite list. To exhaust the remaining signatures without arbitrary length or entry caps, put

$$\mathcal{S}_B = \{a : |a| \geq 2, a_1 > 1, a_1 + 2|a| \leq B\}.$$

Every non-special signature belongs to $\mathcal{S}_B$ for all sufficiently large B . For each B , we contract the strongly connected components of the strict comparison graph and use the deterministic topological order described in the analysis code. At the stated numerical screening precision, the first 69 entries are identical for $B = 18$, $B = 19$, and $B = 20$; Table 2 records this stabilized cutoff. Its displayed rates are then recomputed with the high-accuracy persistent cache.

Because $\prec$ is not transitive, the resulting numbered list is not a linear extension of every pairwise comparison. In Table 3, a row for x records every earlier numbered signature a for which $x \prec a$, together with the two directed rates. Thus each listed entry satisfies

$$C(x | a) < C(a | x)$$

despite a appearing before x .

Table 2: The stabilized first 69 non-special signatures in the deterministic condensation-DAG order.

n	signature	n	signature	n	signature
1	{2, 1}	24	{6, 6}	47	{9, 8}
2	{2, 2}	25	{7, 1}	48	{9, 9}
3	{3, 1}	26	{7, 2}	49	{10, 1}
4	{2, 1, 1}	27	{7, 3}	50	{10, 2}
5	{3, 2}	28	{7, 4}	51	{10, 3}
6	{2, 2, 1}	29	{7, 5}	52	{10, 4}
7	{3, 3}	30	{7, 6}	53	{10, 5}
8	{2, 1, 1, 1}	31	{7, 7}	54	{10, 6}
9	{2, 2, 2}	32	{8, 1}	55	{10, 7}
10	{4, 1}	33	{8, 2}	56	{10, 8}
11	{4, 2}	34	{8, 3}	57	{10, 9}
12	{4, 3}	35	{8, 4}	58	{10, 10}
13	{4, 4}	36	{8, 5}	59	{11, 1}
14	{5, 1}	37	{8, 6}	60	{11, 2}
15	{5, 2}	38	{8, 7}	61	{11, 3}
16	{5, 3}	39	{8, 8}	62	{11, 4}
17	{5, 4}	40	{9, 1}	63	{11, 5}
18	{5, 5}	41	{9, 2}	64	{11, 6}
19	{6, 1}	42	{9, 3}	65	{11, 7}
20	{6, 2}	43	{9, 4}	66	{11, 8}
21	{6, 3}	44	{9, 5}	67	{11, 9}
22	{6, 4}	45	{9, 6}	68	{11, 10}
23	{6, 5}	46	{9, 7}	69	{11, 11}

Table 3: Backward comparisons that obstruct a linear order.

n	x	exceptions with previous n_a values
14	{5, 1}	$n_a = 13, a = \{4, 4\}$: $C(x a) = 0.844794, C(a x) = 0.861353$
19	{6, 1}	$n_a = 16, a = \{5, 3\}$: $C(x a) = 0.897932, C(a x) = 0.898244$ $n_a = 17, a = \{5, 4\}$: $C(x a) = 0.857550, C(a x) = 0.898244$ $n_a = 18, a = \{5, 5\}$: $C(x a) = 0.825066, C(a x) = 0.898244$

continued on the next page

Table 3 continued from the previous page

n	x	exceptions with previous n_a values
25	$\{7, 1\}$	$n_a = 21, a = \{6, 3\}$: $C(x a) = 0.902627, C(a x) = 0.920782$ $n_a = 22, a = \{6, 4\}$: $C(x a) = 0.866396, C(a x) = 0.920782$ $n_a = 23, a = \{6, 5\}$: $C(x a) = 0.837432, C(a x) = 0.920782$ $n_a = 24, a = \{6, 6\}$: $C(x a) = 0.813202, C(a x) = 0.920782$
26	$\{7, 2\}$	$n_a = 23, a = \{6, 5\}$: $C(x a) = 0.906054, C(a x) = 0.920782$ $n_a = 24, a = \{6, 6\}$: $C(x a) = 0.878094, C(a x) = 0.920782$
32	$\{8, 1\}$	$n_a = 24, a = \{6, 6\}$: $C(x a) = 0.844632, C(a x) = 0.861654$ $n_a = 27, a = \{7, 3\}$: $C(x a) = 0.906390, C(a x) = 0.935785$ $n_a = 28, a = \{7, 4\}$: $C(x a) = 0.873076, C(a x) = 0.935785$ $n_a = 29, a = \{7, 5\}$: $C(x a) = 0.846552, C(a x) = 0.935785$ $n_a = 30, a = \{7, 6\}$: $C(x a) = 0.824430, C(a x) = 0.935785$ $n_a = 31, a = \{7, 7\}$: $C(x a) = 0.805397, C(a x) = 0.935785$
33	$\{8, 2\}$	$n_a = 29, a = \{7, 5\}$: $C(x a) = 0.911301, C(a x) = 0.935785$ $n_a = 30, a = \{7, 6\}$: $C(x a) = 0.886261, C(a x) = 0.935785$ $n_a = 31, a = \{7, 7\}$: $C(x a) = 0.864389, C(a x) = 0.935785$
34	$\{8, 3\}$	$n_a = 30, a = \{7, 6\}$: $C(x a) = 0.930247, C(a x) = 0.935785$ $n_a = 31, a = \{7, 7\}$: $C(x a) = 0.906390, C(a x) = 0.935785$

continued on the next page

Table 3 continued from the previous page

n	x	exceptions with previous n_a values
40	$\{9, 1\}$	$n_a = 29, a = \{7, 5\}$: $C(x a) = 0.871798, C(a x) = 0.885622$ $n_a = 30, a = \{7, 6\}$: $C(x a) = 0.850314, C(a x) = 0.885622$ $n_a = 31, a = \{7, 7\}$: $C(x a) = 0.831776, C(a x) = 0.885622$ $n_a = 34, a = \{8, 3\}$: $C(x a) = 0.909518, C(a x) = 0.946395$ $n_a = 35, a = \{8, 4\}$: $C(x a) = 0.878388, C(a x) = 0.946395$ $n_a = 36, a = \{8, 5\}$: $C(x a) = 0.853673, C(a x) = 0.946395$ $n_a = 37, a = \{8, 6\}$: $C(x a) = 0.833105, C(a x) = 0.946395$ $n_a = 38, a = \{8, 7\}$: $C(x a) = 0.815439, C(a x) = 0.946395$ $n_a = 39, a = \{8, 8\}$: $C(x a) = 0.799923, C(a x) = 0.946395$
41	$\{9, 2\}$	$n_a = 35, a = \{8, 4\}$: $C(x a) = 0.942478, C(a x) = 0.946395$ $n_a = 36, a = \{8, 5\}$: $C(x a) = 0.915285, C(a x) = 0.946395$ $n_a = 37, a = \{8, 6\}$: $C(x a) = 0.892295, C(a x) = 0.946395$ $n_a = 38, a = \{8, 7\}$: $C(x a) = 0.872314, C(a x) = 0.946395$ $n_a = 39, a = \{8, 8\}$: $C(x a) = 0.854578, C(a x) = 0.946395$
42	$\{9, 3\}$	$n_a = 37, a = \{8, 6\}$: $C(x a) = 0.933809, C(a x) = 0.946395$ $n_a = 38, a = \{8, 7\}$: $C(x a) = 0.912294, C(a x) = 0.946395$ $n_a = 39, a = \{8, 8\}$: $C(x a) = 0.892960, C(a x) = 0.946395$
43	$\{9, 4\}$	$n_a = 38, a = \{8, 7\}$: $C(x a) = 0.944850, C(a x) = 0.946395$ $n_a = 39, a = \{8, 8\}$: $C(x a) = 0.924272, C(a x) = 0.946395$

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Table 3 continued from the previous page

n	x	exceptions with previous n_a values
49	$\{10, 1\}$	$n_a = 35, a = \{8, 4\}$: $C(x a) = 0.899143, C(a x) = 0.903090$ $n_a = 36, a = \{8, 5\}$: $C(x a) = 0.875102, C(a x) = 0.903090$ $n_a = 37, a = \{8, 6\}$: $C(x a) = 0.855010, C(a x) = 0.903090$ $n_a = 38, a = \{8, 7\}$: $C(x a) = 0.837714, C(a x) = 0.903090$ $n_a = 39, a = \{8, 8\}$: $C(x a) = 0.822502, C(a x) = 0.903090$ $n_a = 41, a = \{9, 2\}$: $C(x a) = 0.952333, C(a x) = 0.954243$ $n_a = 42, a = \{9, 3\}$: $C(x a) = 0.912182, C(a x) = 0.954243$ $n_a = 43, a = \{9, 4\}$: $C(x a) = 0.882762, C(a x) = 0.954243$ $n_a = 44, a = \{9, 5\}$: $C(x a) = 0.859454, C(a x) = 0.954243$ $n_a = 45, a = \{9, 6\}$: $C(x a) = 0.840087, C(a x) = 0.954243$ $n_a = 46, a = \{9, 7\}$: $C(x a) = 0.823475, C(a x) = 0.954243$ $n_a = 47, a = \{9, 8\}$: $C(x a) = 0.808903, C(a x) = 0.954243$ $n_a = 48, a = \{9, 9\}$: $C(x a) = 0.795900, C(a x) = 0.954243$
50	$\{10, 2\}$	$n_a = 38, a = \{8, 7\}$: $C(x a) = 0.894640, C(a x) = 0.903090$ $n_a = 39, a = \{8, 8\}$: $C(x a) = 0.877498, C(a x) = 0.903090$ $n_a = 43, a = \{9, 4\}$: $C(x a) = 0.943960, C(a x) = 0.954243$ $n_a = 44, a = \{9, 5\}$: $C(x a) = 0.918479, C(a x) = 0.954243$ $n_a = 45, a = \{9, 6\}$: $C(x a) = 0.897026, C(a x) = 0.954243$ $n_a = 46, a = \{9, 7\}$: $C(x a) = 0.878445, C(a x) = 0.954243$ $n_a = 47, a = \{9, 8\}$: $C(x a) = 0.862002, C(a x) = 0.954243$ $n_a = 48, a = \{9, 9\}$: $C(x a) = 0.847215, C(a x) = 0.954243$

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Table 3 continued from the previous page

n	x	exceptions with previous n_a values
51	$\{10, 3\}$	$n_a = 45, a = \{9, 6\}$: $C(x a) = 0.936518, C(a x) = 0.954243$ $n_a = 46, a = \{9, 7\}$: $C(x a) = 0.916673, C(a x) = 0.954243$ $n_a = 47, a = \{9, 8\}$: $C(x a) = 0.898942, C(a x) = 0.954243$ $n_a = 48, a = \{9, 9\}$: $C(x a) = 0.882860, C(a x) = 0.954243$
52	$\{10, 4\}$	$n_a = 46, a = \{9, 7\}$: $C(x a) = 0.947468, C(a x) = 0.954243$ $n_a = 47, a = \{9, 8\}$: $C(x a) = 0.928787, C(a x) = 0.954243$ $n_a = 48, a = \{9, 9\}$: $C(x a) = 0.911656, C(a x) = 0.954243$
53	$\{10, 5\}$	$n_a = 48, a = \{9, 9\}$: $C(x a) = 0.936626, C(a x) = 0.954243$

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Table 3 continued from the previous page

n	x	exceptions with previous n_a values
59	$\{11, 1\}$	$n_a = 38, a = \{8, 7\}$: $C(x a) = 0.856507, C(a x) = 0.867194$ $n_a = 39, a = \{8, 8\}$: $C(x a) = 0.841648, C(a x) = 0.867194$ $n_a = 43, a = \{9, 4\}$: $C(x a) = 0.900788, C(a x) = 0.916314$ $n_a = 44, a = \{9, 5\}$: $C(x a) = 0.878004, C(a x) = 0.916314$ $n_a = 45, a = \{9, 6\}$: $C(x a) = 0.859006, C(a x) = 0.916314$ $n_a = 46, a = \{9, 7\}$: $C(x a) = 0.842681, C(a x) = 0.916314$ $n_a = 47, a = \{9, 8\}$: $C(x a) = 0.828343, C(a x) = 0.916314$ $n_a = 48, a = \{9, 9\}$: $C(x a) = 0.815542, C(a x) = 0.916314$ $n_a = 50, a = \{10, 2\}$: $C(x a) = 0.952868, C(a x) = 0.960253$ $n_a = 51, a = \{10, 3\}$: $C(x a) = 0.914492, C(a x) = 0.960253$ $n_a = 52, a = \{10, 4\}$: $C(x a) = 0.886455, C(a x) = 0.960253$ $n_a = 53, a = \{10, 5\}$: $C(x a) = 0.864279, C(a x) = 0.960253$ $n_a = 54, a = \{10, 6\}$: $C(x a) = 0.845875, C(a x) = 0.960253$ $n_a = 55, a = \{10, 7\}$: $C(x a) = 0.830106, C(a x) = 0.960253$ $n_a = 56, a = \{10, 8\}$: $C(x a) = 0.816285, C(a x) = 0.960253$ $n_a = 57, a = \{10, 9\}$: $C(x a) = 0.803964, C(a x) = 0.960253$ $n_a = 58, a = \{10, 10\}$: $C(x a) = 0.792835, C(a x) = 0.960253$

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Table 3 continued from the previous page

n	x	exceptions with previous n_a values
60	$\{11, 2\}$	$n_a = 45, a = \{9, 6\}$: $C(x a) = 0.915642, C(a x) = 0.916314$ $n_a = 46, a = \{9, 7\}$: $C(x a) = 0.897629, C(a x) = 0.916314$ $n_a = 47, a = \{9, 8\}$: $C(x a) = 0.881636, C(a x) = 0.916314$ $n_a = 48, a = \{9, 9\}$: $C(x a) = 0.867225, C(a x) = 0.916314$ $n_a = 52, a = \{10, 4\}$: $C(x a) = 0.945256, C(a x) = 0.960253$ $n_a = 53, a = \{10, 5\}$: $C(x a) = 0.921131, C(a x) = 0.960253$ $n_a = 54, a = \{10, 6\}$: $C(x a) = 0.900885, C(a x) = 0.960253$ $n_a = 55, a = \{10, 7\}$: $C(x a) = 0.883391, C(a x) = 0.960253$ $n_a = 56, a = \{10, 8\}$: $C(x a) = 0.867946, C(a x) = 0.960253$ $n_a = 57, a = \{10, 9\}$: $C(x a) = 0.854085, C(a x) = 0.960253$ $n_a = 58, a = \{10, 10\}$: $C(x a) = 0.841485, C(a x) = 0.960253$
61	$\{11, 3\}$	$n_a = 48, a = \{9, 9\}$: $C(x a) = 0.902904, C(a x) = 0.916314$ $n_a = 53, a = \{10, 5\}$: $C(x a) = 0.959953, C(a x) = 0.960253$ $n_a = 54, a = \{10, 6\}$: $C(x a) = 0.938697, C(a x) = 0.960253$ $n_a = 55, a = \{10, 7\}$: $C(x a) = 0.920119, C(a x) = 0.960253$ $n_a = 56, a = \{10, 8\}$: $C(x a) = 0.903587, C(a x) = 0.960253$ $n_a = 57, a = \{10, 9\}$: $C(x a) = 0.888649, C(a x) = 0.960253$ $n_a = 58, a = \{10, 10\}$: $C(x a) = 0.874982, C(a x) = 0.960253$

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Table 3 continued from the previous page

n	x	exceptions with previous n_a values
62	$\{11, 4\}$	$n_a = 55, a = \{10, 7\}$: $C(x a) = 0.949460, C(a x) = 0.960253$ $n_a = 56, a = \{10, 8\}$: $C(x a) = 0.932147, C(a x) = 0.960253$ $n_a = 57, a = \{10, 9\}$: $C(x a) = 0.916370, C(a x) = 0.960253$ $n_a = 58, a = \{10, 10\}$: $C(x a) = 0.901830, C(a x) = 0.960253$
63	$\{11, 5\}$	$n_a = 56, a = \{10, 8\}$: $C(x a) = 0.956636, C(a x) = 0.960253$ $n_a = 57, a = \{10, 9\}$: $C(x a) = 0.940214, C(a x) = 0.960253$ $n_a = 58, a = \{10, 10\}$: $C(x a) = 0.924929, C(a x) = 0.960253$
64	$\{11, 6\}$	$n_a = 58, a = \{10, 10\}$: $C(x a) = 0.945667, C(a x) = 0.960253$

B.1 The global comparison clusters

On the infinite family of non-special integer signatures, we distinguish a non-strict and a strict comparison arrow:

$$a \longrightarrow b \iff C(a|b) \geq C(b|a), \quad a \xrightarrow{s} b \iff C(a|b) > C(b|a).$$

Thus an arrow points from the greater element toward the smaller one, like the $>$ symbol; equivalently, $a \xrightarrow{s} b$ means $b \prec a$. Equality produces two opposite non-strict arrows and no strict arrow. Define the corresponding clusters as the strongly connected components

$$\text{Cl}_{\geq}(a) = \{b : a \longrightarrow^* b \text{ and } b \longrightarrow^* a\},$$

and analogously define $\text{Cl}_{>}(a)$ using only $\xrightarrow{s}$. Reversing the orientation used for the arrows does not change either strongly connected component.

This is an infinite-graph question: a strongly connected component of a finite induced subgraph need not equal the intersection of the global component with that subgraph, because paths may pass through signatures outside the truncation. In particular, the 69-prefix give misleadingly small components.

Calculations on the nested finite shells $\mathcal{S}_B$ suggest the following non-strict global description, which we leave as a conjecture:

$$\text{Cl}_{\geq}(\{3, 1, 1\}) = \{a : |a| \geq 2, a_1 > 1\} \setminus \{\{2, 1\}, \{2, 2\}\}.$$

For example, the component contains 112 of the 129 signatures in $\mathcal{S}_{12}$, and 2498 of the 2560 signatures in $\mathcal{S}_{18}$. Across the tested shells $12 \leq B \leq 18$, these finite components do not stabilize: signatures near the boundary of a smaller shell join through paths appearing in larger shells. Thus no finite list is asserted here.

The strict cluster appears to have an equally simple description. Put

$$\mathcal{E}_s = \{ \{2, 1\}, \{2, 2\}, \{3, 1\}, \{3, 2\}, \{3, 3\}, \{4, 1\}, \{4, 2\}, \{4, 3\}, \\ \{2, 1, 1\}, \{2, 2, 1\}, \{2, 2, 2\}, \{2, 1, 1, 1\} \}.$$

Our strict-cluster conjecture is

$$\text{Cl}_>(\{3, 1, 1\}) = \{a : |a| \geq 2, a_1 > 1\} \setminus \mathcal{E}_s.$$

As finite-shell evidence, the strict component in $\mathcal{S}_{20}$ contains 6719 of its 6738 signatures. Its intersection with $\mathcal{S}_{19}$ is exactly $\mathcal{S}_{19} \setminus \mathcal{E}_s$; the remaining seven excluded signatures are all new boundary points with $a_1 + 2|a| = 20$. This pattern is evidence rather than a proof that later shells absorb every boundary exception and never absorb a member of $\mathcal{E}_s$.

Here is one explicit strict cycle, computed inside $\mathcal{S}_{14}$ with comparison tolerance $2 \cdot 10^{-6}$. The shortest strict path from $\{9, 1\}$ to $\{3, 2, 2, 2\}$, in the arrow orientation above, is

$$\{9, 1\} \xrightarrow{s} \{3, 3, 3\} \xrightarrow{s} \{3, 1, 1, 1\} \xrightarrow{s} \{8, 4\} \xrightarrow{s} \{10, 1\} \xrightarrow{s} \{3, 2, 2, 2\}.$$

Every row of Table 4 satisfies $C(a \mid b) > C(b \mid a)$.

Table 4: Strict path from $\{9, 1\}$ to $\{3, 2, 2, 2\}$.

Strict edge $a \xrightarrow{s} b$	$C(a \mid b)$	$C(b \mid a)$
$\{9, 1\} \xrightarrow{s} \{3, 3, 3\}$	0.630929754	0.500000000
$\{3, 3, 3\} \xrightarrow{s} \{3, 1, 1, 1\}$	0.792481250	0.753940561
$\{3, 1, 1, 1\} \xrightarrow{s} \{8, 4\}$	0.527921767	0.500000000
$\{8, 4\} \xrightarrow{s} \{10, 1\}$	0.903089987	0.899142666
$\{10, 1\} \xrightarrow{s} \{3, 2, 2, 2\}$	0.500000000	0.477121255

The shortest strict return path has only two edges:

$$\{3, 2, 2, 2\} \xrightarrow{s} \{7, 5\} \xrightarrow{s} \{9, 1\}.$$

Table 5: Strict return path from $\{3, 2, 2, 2\}$ to $\{9, 1\}$.

Strict edge $a \xrightarrow{s} b$	$C(a \mid b)$	$C(b \mid a)$
$\{3, 2, 2, 2\} \xrightarrow{s} \{7, 5\}$	0.564574819	0.500000000
$\{7, 5\} \xrightarrow{s} \{9, 1\}$	0.885621875	0.871797909

B.2 A length-three strict cycle

Strict cyclicity is not caused by comparing signatures of different lengths. An exhaustive high-accuracy evaluation of the 55 non-special length-three signatures with entries at most 6 finds no strict three-cycle at comparison tolerance $2 \cdot 10^{-6}$. The first maximum entry for which the bounded search finds one is 7:

$$\{6, 3, 3\} \xrightarrow{s} \{7, 2, 1\} \xrightarrow{s} \{6, 5, 1\} \xrightarrow{s} \{6, 3, 3\}.$$

Table 6 gives both directed rates on every edge.

Table 6: A strict three-cycle of length-three signatures.

Strict edge $a \xrightarrow{s} b$	$C(a \mid b)$	$C(b \mid a)$	difference
$\{6, 3, 3\} \xrightarrow{s} \{7, 2, 1\}$	0.920782221	0.910910734	0.009871488
$\{7, 2, 1\} \xrightarrow{s} \{6, 5, 1\}$	0.924704131	0.920782221	0.003921910
$\{6, 5, 1\} \xrightarrow{s} \{6, 3, 3\}$	0.975533650	0.959474358	0.016059292